

not even half as good as original 3D content.

For one, not all content looks good in 3D. 3D is best suited for certain types of content, such as slow panoramic shots, and does not work with all kinds of imagery. Hence a lot of jolting and crude transitions are introduced into the converted movie—which certainly doesn't give a pleasant result. Moreover, the quality of conversion is dependent on too many factors, ranging from the type and quality of movie to the type of equipment on which it is going to be telecast, making it a very unpredictable business.

Nevertheless, it is an option worth exploring. If perfected, it could provide an unending source of 3D content, and a great motivation to buy a 3D television. Plus, experts also feel that this is a technique cable operators will prefer to pick up, because it puts less pressure on their data networks. They would be transmitting only traditional 2D content to the users' home, leaving their 3D set-top boxes to do the conversion. On the other hand, transmitting real 3D content would mean sending more information across their data networks, requiring upgrades to stand the load.

Samsung was the first company to introduce a model with real-time 2D-to-3D conversion. Other brands have followed suit.

Need for standardisation

"There are many challenges. First is standardisation of broadcasting norms and specifications. Second is to make it glassless viewing. Standardisation is around the corner, while glassless viewing will take some time. But even with glasses, it is very exciting and acceptable viewing," says Sharma.

Broadcasting is just one of the areas where the need for standardisation is greatly felt. The technology behind glasses, disk players, bandwidth considerations, recording formats, and almost every aspect of 3D television is much in need of some uniformity and interoperability. You can find a lot of



funny but true stories on the Internet, such as how one brand's 3D glasses worked with another brand's TV only when worn upside down.

As it happens during every disruptive technological advancement, 3D technology is also sure to be torn in a battle of formats, before we see some standardisation. Almost every major standards body has a 3D standard to offer.

The Motion Picture Experts Group (MPEG) has come out with the multi-view video coding (MVC) extension for MPEG-4 AVC, which enables efficient encoding of sequences captured simultaneously from multiple cameras using a single video stream. It can be used for encoding stereoscopic (two-view) video, as well as free viewpoint television and multi-view 3D television. The 3D version of Blu-ray will be using the MVC, which is also backwards-compatible with 2D Blu-ray players.

High-definition Multimedia Interface (HDMI) version 1.4 also defines a number of 3D transmission formats, and HDMI 1.4 devices will be capable of transmitting 3D pictures in full 1080p resolution.

The Digital Video Broadcasting (DVB) Project, for its part, has established the DVB 3D-TV specification. It

supports various configurations such as a 3D TV connected to a 3D Blu-ray player for packaged media, a 3D TV connected to an HD gaming console like the PS3 for 3D gaming, a 3D TV connected to an HD set-top box for broadcast 3D content, as well as a 3D TV receiving 3D TV broadcast directly via a built-in tuner and decoder.

The International Telecommunications Union (ITU), the Society of Motion Picture and Television Engineers, the Consumer Electronics Association and the Advanced Television Systems Committee in the United States are also working towards standardisation of 3D technologies. The ITU has announced that 3D TV would eventually roll out in three gradual stages: first with dual-image stereoscopy, then with multi-view systems, and finally a third generation that will feature systems which record the amplitude, frequency and phase of light waves to reproduce almost completely human beings' natural viewing environment.

Picking a New Year gift?

It might be too premature to buy yourself a 3D television this New Year, but next year you might be considering it quite seriously. Despite the content crunch and the dire need for standardi-